

# Air, Ventilation & Environment

Aangan · Nathaniel School of Music · 30 Jul 2026 · what the Dyson can already do, what to add per room, and the automations that make it a system rather than a shelf of gadgets

**Short answer:** yes to all three. (1) Your existing Dyson can join Aangan for ₹0 — it has excellent sensors inside it that we're currently wasting. (2) For other rooms, do **not** buy Dyson again — Xiaomi purifiers integrate natively for about a quarter of the price. (3) The Pi doesn't purify air; it **decides** — and that decision layer (recording-aware, class-aware, CO<sub>2</sub>, odour, humidity) is worth more than any single purifier.

## 1 · The Dyson you already own ₹0 · the biggest win in this document

Wi-Fi Dyson purifiers speak MQTT, and Home Assistant talks to them through the community **ha-dyson** integration (installed via HACS). After a one-time sign-in it runs **fully locally** — no cloud in the daily loop.

### What we gain immediately

- **Its sensors become the house's sensors.** The Dyson already measures PM2.5, PM10, VOCs, NO<sub>2</sub>, temperature and humidity to a standard better than most DIY parts. That data flows straight into Aangan's House Pulse — a professional air sensor we already paid for and currently can't see.
- **Full control:** fan speed, auto mode, night mode, oscillation, plus filter life remaining (so the app can say “order a filter” instead of you discovering it).
- **It joins the studio logic.** This is the real prize — see the automations below.

**One honest caveat, resolved in five minutes:** Dyson removed the local Wi-Fi credential sticker on models from roughly mid-2021 onward, so those need a single one-time MyDyson cloud login for the integration to fetch credentials — after that it's local. Very recent models occasionally lag support. **Send me the model number from the sticker under the base** (e.g. TP07, HP09, PH04) and I'll confirm support before we count on it. Cost either way: nothing.

## 2 · Purifiers for the other rooms the affordable answer

A second Dyson is roughly ₹45,000–55,000. It is a superb machine and it belongs where you record and teach — which is where you already have it. For bedrooms and the teaching room, the sensible buy is Xiaomi: **native Home Assistant support** (the built-in Xiaomi integration, local control), real HEPA + activated-carbon filtration, and about a quarter of the price.

| ROOM                | WHAT GOES THERE                          | WHY                                                                              | REF. PRICE |
|---------------------|------------------------------------------|----------------------------------------------------------------------------------|------------|
| Music room / studio | the Dyson you own ₹0                     | best filtration + best sensors, already there; becomes recording-aware           | ₹0         |
| Bedroom (Amma)      | Xiaomi Smart Air Purifier 4<br>BUY FIRST | where a person breathes 8 hours a night — the highest health return in the house | ~₹13,000   |

|                              |                                  |                                                                    |             |
|------------------------------|----------------------------------|--------------------------------------------------------------------|-------------|
| <b>Teaching room</b>         | Xiaomi Smart Air Purifier 4 Lite | students in a closed room; runs max between classes, silent during | ~₹10,000    |
| <b>Other bedrooms / hall</b> | Xiaomi 4 Lite, added later       | scale only after the first unit's automation is proven             | ~₹10,000 ea |

**Do not use the smart-plug trick on a purifier.** Cutting power to a “dumb” purifier looks clever, but most units do not resume their previous speed when power returns — some need a physical button press, so the house silently stops purifying and nobody notices. Buy units the system can actually talk to (Dyson / Xiaomi). The one appliance where a plug or relay IS correct is a simple exhaust fan, which has no state to lose.

### 3 · The sensing layer where the Raspberry Pi earns its place · ~₹6–8k per room

A purifier only knows the air at its own grille, and only reacts. The Pi's job is to know the whole flat and act *ahead* of the problem. Each air node is one ESP32 (six already owned) plus these sensors, reporting to the House Pi over Wi-Fi:

| SENSOR                                                | MEASURES               | WHY IT MATTERS HERE                                                                                                                                                                                                                                               |
|-------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PMS5003</b>                                        | PM2.5 / PM10 dust      | Bangalore road dust, and dust that visibly shows up in video                                                                                                                                                                                                      |
| <b>SCD41</b> (true CO <sub>2</sub> , not an estimate) | CO <sub>2</sub> ppm    | <b>The one everybody misses.</b> A soundproofed room with the doors shut and four people in it climbs past 1,400 ppm fast — that is exactly why students get dull and heads get heavy in the second half of a class. You cannot smell it. Only a sensor finds it. |
| <b>SGP41</b>                                          | VOC index + NOx index  | This <i>is</i> the “smell” number — a live staleness/odour index. Cooking, paint, damp, bathroom, a closed room after a long day.                                                                                                                                 |
| <b>SHT45</b> (or the owned DHT22)                     | temperature / humidity | mould risk in monsoon <b>and instrument protection</b> — see \$5                                                                                                                                                                                                  |

**Where to put the three nodes:** ① music room / studio (the room that earns money) ② kitchen (cooking smoke, odour, and it doubles as backup to the gas sensors) ③ Amma's bedroom (night-time CO<sub>2</sub> and humidity).

### 4 · The automations — this is the actual product

Gadgets on their own get ignored within a month. These rules are what make it a system, and every one of them is a rule the app can show and you can override.

| AUTOMATION                         | WHAT HAPPENS                                                                                                                                                                                                                                                                                                                | NEEDS             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Recording-aware purifying ★</b> | A purifier is a fan — it is noise on the take. When the dial goes Audio Rec or Video Rec, the Dyson drops to silent or off and joins the existing “Silence the room” pre-flight; when the studio returns to Available it goes back to purifying at full. Right now this is a manual thing you have to remember mid-session. | Dyson integration |

|                                            |                                                                                                                                                                                                                                                                                                           |                                                          |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Pre-class / pre-take purge ★</b>        | Twenty minutes before a calendar class or a shoot, every purifier runs on maximum; the moment the class starts they go quiet. Students walk into clean, silent air.                                                                                                                                       | Dyson + calendar (already built)                         |
| <b>CO<sub>2</sub> ventilation nudge ★</b>  | Studio or bedroom CO <sub>2</sub> over ~1,000 ppm → “open a window / crack the door” on the wall panels and phones. During a class, over ~1,400 ppm it nudges harder. A purifier does <b>not</b> fix CO <sub>2</sub> — only fresh air does, which is why this is a ventilation rule, not a purifier rule. | SCD41 node                                               |
| <b>Kitchen odour auto-exhaust</b>          | Odour index or cooking dust spikes → the kitchen exhaust fan runs for N minutes and stops itself. It will not start during a take (that's noise) — <b>unless</b> gas is detected, in which case safety always wins and it runs regardless.                                                                | SGP41 node + fan relay<br><a href="#">electrician</a>    |
| <b>Bathroom damp control</b>               | Humidity jumps after a shower → exhaust fan runs until the room is dry again. Ends the “someone forgot the fan, now there's mould” cycle.                                                                                                                                                                 | humidity node + fan relay<br><a href="#">electrician</a> |
| <b>Instrument climate guard ★</b>          | Any room holding above ~65% RH (or below ~35%) for hours → alert, because that is what warps pianos, guitars and bows. See §5 — for a music school this may be the single most valuable rule in the list.                                                                                                 | humidity nodes                                           |
| <b>Air row in Pre-flight</b>               | The go/no-go verdict gains an air line: don't start a shoot with dust high — it shows on camera and means the room wants purging first.                                                                                                                                                                   | any node                                                 |
| <b>Purifier running with a window open</b> | The door/window sensors already exist — if a purifier is working hard while a window is wide open, the app says so instead of quietly wasting the filter.                                                                                                                                                 | reeds (on order)                                         |
| <b>Filter life &amp; reorder</b>           | Dyson and Xiaomi both report filter life; Aangan warns at 10% instead of you finding out from a red light.                                                                                                                                                                                                | purifier integrations                                    |

★ = the four I would build first. Recording-aware purifying and the pre-class purge pay for themselves in your working day; the CO<sub>2</sub> nudge fixes an invisible problem that affects how well students actually learn; the instrument guard protects assets worth far more than the sensors.

## 5 · “Anything for that matter” — the wider environment layer

### The one I would insist on: instrument climate

Pianos want roughly 40–60% relative humidity and hate swings. Bangalore's monsoon pushes indoor humidity well past that, and the studio is sealed. Two humidity sensors and a rule that shouts early is insurance on instruments worth lakhs — and if it turns out a room sits damp for weeks, the fix is a dehumidifier on a relay (dehumidifiers, unlike purifiers, do resume after power returns — we verify the specific model first).

### Also worth doing, cheap

- **RO water quality** — a TDS probe on the drinking line (~₹600) tells you the RO filter is finished before the water tastes wrong.

- **AC efficiency nudge** — “AC running with the balcony door open” is a rule we can write today from sensors already on order.
- **Circadian light** — the WS2812B strip you own can warm down after sunset; smart bulbs later if you like the effect.
- **Kitchen chimney reminder** — cooking dust spike with no chimney running → a nudge to switch it on.
- **Balcony rain + plants** — the owned rain sensor and soil-moisture sensor already cover “bring the clothes in” and “the planter is dry”.

**Deliberately NOT recommending:** ozone generators or “ionisers” for odour (ozone is a lung irritant — never in an occupied room, and definitely not near singers), and cheap all-in-one “air quality” modules that estimate CO<sub>2</sub> from VOCs instead of measuring it. If we are going to act on a number, the number has to be real.

## 6 · What to buy added to the procurement list as Order 3 — nothing here disturbs Orders 1 and 2

| ITEM                                                               | QTY | REF. PRICE | WHERE                                     |
|--------------------------------------------------------------------|-----|------------|-------------------------------------------|
| Xiaomi Smart Air Purifier 4 — bedroom                              | 1   | ₹13,000    | <a href="https://mi.com/in">mi.com/in</a> |
| Xiaomi Smart Air Purifier 4 Lite — teaching room                   | 1   | ₹10,000    | <a href="https://mi.com/in">mi.com/in</a> |
| PMS5003 particulate sensor                                         | 3   | ₹2,000 ea  | <a href="#">robu / SP Road</a>            |
| SCD41 true CO <sub>2</sub> sensor module                           | 3   | ₹3,500 ea  | <a href="#">robu / SP Road</a>            |
| SGP41 VOC + NOx sensor module                                      | 3   | ₹2,000 ea  | <a href="#">robu / SP Road</a>            |
| SHT45 temperature/humidity module                                  | 2   | ₹1,000 ea  | <a href="#">robu / SP Road</a>            |
| Relay module for exhaust fans <small>fitted by electrician</small> | 3   | ₹250 ea    | SP Road electrical                        |
| TDS water quality probe                                            | 1   | ₹600       | <a href="#">robu / SP Road</a>            |
| Project boxes for the three air nodes                              | 3   | ₹200 ea    | SP Road                                   |

ESP32 boards, DHT22 and the jumper/breadboard/charger bundle are already owned or already on Order 1 — nothing duplicated here.

**Order 3 total ≈ ₹47,000** — and it is genuinely staged: the **₹0 Dyson integration first** (do it the day the House Pi boots and see whether the data alone changes how you run the rooms), then **one Xiaomi + one air node ≈ ₹20,500** to prove the automations in the room that matters most, then scale. If only one line ever gets bought, buy the CO<sub>2</sub> sensor for the studio — it answers a question nothing else in the house can.